

Code-Layout, Readability and Reusability

Date	30 September 2026
Team ID	SWTID-2026-7291
Project Name	Weather-Wise
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. The Weather-Wise backend is organized into routes, controllers, models, middleware, and services.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed / (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Consistent formatting is followed.
2	Proper File Structure	Files and folders are logically organized	Yes	Routes, controllers, models and middleware are separated.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Names are generally clear and descriptive.
4	Function / Method Names	Functions are descriptively named	Yes	Functions reflect their main purpose.
5	Code Comments	Inline and block comments explain logic	Partial	Comments can be expanded for complex sections.
6	Modular Design	Code is split into reusable functions/modules	Yes	Backend is divided into reusable modules.
7	No Redundant Code	Duplicate or unused code is removed	Partial	Final cleanup/review is recommended.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	API and authentication errors are handled.

Reusable Components / Modules:

Aspect	Rating (1-5)	Comments		
1	Authentication Middleware	JavaScript / Express	Protected user and API routes	High
2	Weather Controller / Service	JavaScript / Node.js	Weather API requests	High
3	AI Controller	JavaScript / Node.js	AI recommendation and summary features	Medium
4	API Routes	JavaScript / Express	Authentication, weather and AI endpoints	High
5	MongoDB / Mongoose Models	JavaScript / Mongoose	User data and database operations	High

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Well-organized backend folders and clear separation of responsibilities.
Readability	4	Naming and API structure are clear and easy to follow.
Reusability	4	Authentication, routes and controllers are designed as reusable modules.
Documentation / Comments	3	Basic structure is clear; more comments can be added to complex logic.
Overall Score	4	Good overall structure with minor improvements recommended.

Note: Ratings are based on the Weather-Wise backend structure demonstrated in the project materials.